

Question. (Erdős #741(ii)) Is there a basis A of order 2 such that if $A = A_1 \sqcup A_2$ then $A_1 + A_1$ and $A_2 + A_2$ cannot both have bounded gaps?

Proof. We show that the answer is yes by constructing a pathological basis A with “forbidden zones” that force any partition to create arbitrarily large gaps in at least one component sumset.

Step 1: The construction. Choose a sequence of rapidly growing scales $P_k = 100^k$. For each $k \geq 1$, define a forbidden zone Z_k , which is a broad interval of integers running roughly from $\frac{11}{2}P_k$ to $11P_k + k$. In the middle of this zone, we leave a single “oasis” — an isolated element $x_k = 10P_k$. The set A consists of all natural numbers that avoid every forbidden zone, together with the oases x_k . In visual terms, A is made of clumps of consecutive integers separated by large empty gaps, each containing a single survivor x_k .

Step 2: $A \cup \{0\}$ is a basis of order 2. We must show every natural number n can be written as $a + b$ with $a, b \in A \cup \{0\}$:

- If $n \notin Z_k$ for any k , then $n \in A$ and we use $n + 0 = n$.
- If $n \in Z_k$ and n lies in the lower half of Z_k , we use $\lfloor n/2 \rfloor + \lceil n/2 \rceil$. Because n is small enough, neither half lands in Z_k , and both are large enough to avoid the previous zone Z_{k-1} .
- If $n \in Z_k$ and n lies in the upper half, we use the oasis: $n = x_k + (n - x_k)$. The difference $n - x_k$ is small enough that it falls safely before the forbidden zone Z_k .

Step 3: No syndetic partition exists. Suppose $A = A_1 \sqcup A_2$. Consider a target sum $m \in [11P_k, 11P_k + k]$. To write $m = u + v$ with $u, v \in A$, the algebra forces the larger operand v to land inside the forbidden range $[\frac{11}{2}P_k, 11P_k + k]$. But the only element of A in this range is x_k . Therefore, representing any sum in $[11P_k, 11P_k + k]$ requires x_k .

By the pigeonhole principle, x_k belongs to exactly one partition component, say A_1 . Then $A_1 + A_1$ can cover the interval $[11P_k, 11P_k + k]$ using x_k , but $A_2 + A_2$ is completely locked out — it cannot represent any element in that interval. This leaves a gap of length k in $A_2 + A_2$. Since k can be made arbitrarily large, the gaps in one component’s sumset are unbounded, proving it is impossible to partition A so that both sumsets have bounded gaps. $\square$